

ANSWER KEY AND MARKING RUBRIC

Mock Examination: Answer Key

9DATAV Data Visualisation · MSCBADM · April 2026

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Use this rubric to self-mark after sitting the mock under timed, closed-book conditions. Reading it before attempting the paper will undermine the diagnostic value of the practice.

Overall Philosophy for Markers

This mock mirrors the format and expectations of the May terminal. It is calibrated to be slightly more demanding (more stakeholders, more ambiguity, an explicit AI praxis path) while keeping the same marking standards and grade boundaries.

1. **Reward understanding over recall.** A student who explains a concept accurately in their own words, with a good example, scores as highly as one who cites a textbook definition.
2. **Credit experiential evidence.** Specific references to the DataVis Challenge (dataset, design choices, what worked, what would change) are evidence of applied understanding.
3. **Named principles push into higher bands.** Kirk (2019), Knafllic (2015), Ware (2012), Tufte (2001), Few (2009), Cleveland and McGill (1984), Shneiderman (1996) lift strong answers into 70%+. Absence of citations does not cap a strong answer below 60%.
4. **Reward honesty and reflection.** Acknowledging limitations is Level 9 thinking.
5. **Penalise vagueness, not informality.** "The colours are confusing" earns less than "the rainbow palette is perceptually non-uniform (Ware, 2012), creating false boundaries between adjacent hues."

Target Grade Distribution

- Attended sessions + engaged DataVis Challenge + moderate study: **55-65%** on this mock; likely **60-69%** (2.1) on the terminal.
- Additionally wider reading + strong critical reflection: **65-75%** on this mock; **70%+** (first class) on the terminal.
- Minimal engagement: capable of passing (**40-49%**) with basic understanding.
- Approximate balance: ~35% class experience and DataVis Challenge, ~40% theory study, ~25% wider reading and critical depth.

Section A: Marking Notes

Question A1 (8 marks) : Both paths equivalent

A1(i) Theory. (a) Exploratory ("discovery-oriented, analyst-facing, rapid, iterative") and explanatory ("communication-oriented, audience-facing, polished, single message"). Full marks: precise definitions + Kirk (2019) or Knafllic (2015). 3: correct without source. 2: one mode well, one weak. 0-1: confused.

(b) Two distinct shifts justified. Strong: simplification (Tufte), single message (Knafllic), annotation that states the insight, audience-appropriate chart type. 3: two with one weak. 2: one well-justified. 0-1: only restates exploratory/explanatory difference.

A1(ii) Applied. (a) Specific reference to two named charts in submission with named differences. Full: genuine engagement with accurate framing. 3: charts named, differences generic. 2: vague.

(b) Honest, specific identification of an "explanatory" chart still carrying exploratory residue, with a credible simplification grounded in a principle. 3: honest but thin. 2: defensive.

Question A2 (7 marks) (LO2)

(a) Pre-attentive processing (Ware, 2012); attribute = colour hue. Full: all three (mechanism, attribute, source). 2: two correct. 1: one.

(b) Two further pre-attentive attributes (size, motion, orientation, position, enclosure, intensity) each paired with a credible scenario. Full: two well-justified. 3: scenarios thin. 0-1: attributes are not pre-attentive.

Question A3 (8 marks) (LO1, LO5)

(a) Data-ink ratio (4 marks). Three of: 3D effect, photographic background, drop shadows, redundant percentage labels (slices + legend), excessive bright hues. Each justified. Tufte (2001) reference strengthens. 2: two with thin justification. 1: one. 0: cannot apply.

(b) Cleveland and McGill (4 marks). Pies force angle/area comparison, both ranked far below position along a common scale (Cleveland and McGill, 1984). Six categories make adjacent-slice comparison especially difficult. Recommend bar/dot plot sorted by value, with justification (most accurate perceptual channel). 2: cites hierarchy, recommendation generic. 1: notes pies are bad without theoretical grounding.

Question A4 (7 marks) : Both paths equivalent

A4(i) Theory. (a) (1) Perceptual non-uniformity (Ware, 2012; Borland and Taylor, 2007); (2) No intuitive ordering. Other valid: accessibility for CVD. 3: two issues, one without source. 2: one well, one weak.

(b) Sequential single-hue (e.g. ColorBrewer YlOrRd, Viridis) with justification (perceptual uniformity, intuitive ordering, accessibility) + one accessibility consideration. Full: all three.

A4(ii) AI Praxis. (a) Specific situation. Full: clear flaw using named principles. 3: situation named, plain language. 2: vague.

(b) Names limit (AI optimises for output completeness, lacks audience model) + minimum human review (graphical integrity, perceptual channel, audience-fit, "so what" check). Full: both elements with practical specificity.

Section B: Marking Guidance

B1: DataVis Challenge Reflection (LO1, LO5)

More demanding than terminal: students must interrogate the limits of their own design.

Applied (a) : Most important insight (10 marks)

Band	Descriptor
70%+ (7-10)	Names insight specifically. Honestly evaluates novelty (surprising or confirmatory?). Honesty itself is critical thinking. Distinguishes "finding" from "insight".
60-69% (6)	Identifies insight with reasonable specificity; some evaluation of value.
50-59% (5)	Describes finding without evaluating value.
40-49% (4)	Generic description.
0-39% (0-3)	Cannot identify a meaningful insight.

Applied (b) : Reversed design decision (15 marks)

Band	Descriptor
70%+ (11-15)	Specific decision named. Justification grounded in a named principle (e.g. "stacked area chart for time series across 5 categories; Cleveland and McGill rank colour-area below position; would now use small-multiple line chart"). Honest, reflective, specific.
60-69% (9-10)	Decision named, justification reasonable, one principle cited.
50-59% (7-8)	Decision named; justification thin or generic ("change the colours").
40-49% (6)	Defensive or vague.
0-39% (0-5)	Cannot identify a reversal.

Applied (c) : Data vs claim gap (10 marks)

Band	Descriptor
70%+ (7-10)	Engages with implied causality where data only supports correlation; sample limitations not surfaced; uncertainty hidden by clean lines or sharp boundaries; persuasive effect of polished design even on preliminary analysis. Connects to ethics of visualisation.
60-69% (6)	Good reflection, one ethical connection.
50-59% (5)	Surface-level.

0-39% (0-4)	Minimal or no reflection.
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Theory Path uses same band structure. Look for: (a) genuine engagement with "every visualisation is editorial" (choice of inclusion, framing, chart type, palette are all editorial); (b) graphical integrity violation by omission rather than distortion, with discussion of why omission is harder to detect (no visible axis truncation, no obvious lie factor; deception invisible at chart surface); (c) named ethical duties: honesty in framing, methodology disclosure, asymmetry of expertise.

B2: Iverna Health Trust (LO2, LO5)

More complex than terminal's GreenLine: clinical safety stakes (missed sepsis alerts), specific design fault (critical panel in low-attention zone with low-saturation colour), validated redesign required.

(a) Evaluation (15 marks)

Band	Descriptor
70%+ (11-15)	Three+ mechanisms with depth. Cognitive load: 16 panels exceed working memory (Miller; Ware). Pre-attentive failure: pale teal of sepsis panel does not pop against three matching panels (Ware). Visual hierarchy violation: critical metric in bottom-right (F/Z reading patterns; Few). Figure-ground: dense layout, no quiet zones. Decorative or undifferentiated colour wastes a high-value channel. Clinical safety connection strengthens.
60-69% (9-10)	Three mechanisms, good depth on at least two.
50-59% (7-8)	Two-three mechanisms, generic without theoretical grounding.
40-49% (6)	Surface-level.
0-39% (0-5)	Fails to engage with perception/cognition theory.

(b) Redesign (20 marks)

Band	Descriptor
70%+ (14-20)	Primary zone top-left for sepsis with high-saturation pre-attentive cue (size, distinct hue, motion or border for active alerts). Secondary panels demoted via progressive disclosure (named tabs, expand-on-demand). Tertiary (weather, news) demoted further or moved to separate view. Clinical staff involved through workshops, paper prototyping, shift walk-throughs. Validation: simulation, eye-tracking, shift shadowing. Shneiderman, Few, user-centred design references strengthen.
60-69% (12-13)	All five elements; good depth on three.
50-59% (10-11)	All five but generic; progressive disclosure mentioned without development.
40-49% (8-9)	At least three elements, vague.
0-39% (0-7)	Absent or incoherent.

B3: Atlas FoodCo (LO1, LO2, LO5)

Tests audience-specific design (like terminal's RetailPulse) but adds a specific bad recommendation ("one dashboard for everyone with filters") that the student must dismantle, not merely critique.

(a) Evaluation of one-dashboard proposal (15 marks)

Band	Descriptor
70%+ (11-15)	Specific cognitive/design problems for each audience. CEO: drowns in operational detail. Operations director: filtered views miss bottleneck-relevant chart types and refresh cadence. Category managers: filters cannot substitute for SKU-level conditional formatting and dense drill-down. Cognitive load (Miller; Ware), audience-first design (Kirk; Knafllic), Few's emphasis. Layered architecture (shared semantic model + audience-specific presentation) recommended with justification.
60-69% (9-10)	All three audiences addressed; principles named; one weaker.
50-59% (7-8)	Two audiences, generic ("they need different things").
40-49% (6)	One audience or no theoretical grounding.
0-39% (0-5)	Cannot dismantle the proposal.

(b) Audience-specific recommendations (20 marks)

Band	Descriptor
70%+ (14-20)	CEO: summary KPIs, sparklines, minimal interactivity (Tufte minimalism, Knafllic single-message). Ops director: comparative views, small multiples, drill-down, weekly cadence (Few exception highlighting; Shneiderman zoom-and-filter). Category managers: dense grid with conditional formatting, high interactivity, daily refresh, SKU filters (Shneiderman details-on-demand; Cleveland and McGill position channel). Each recommendation justified with named principle.
60-69% (12-13)	All three with named chart types and one principle each.
50-59% (10-11)	All three but undifferentiated or unjustified.
40-49% (8-9)	Two audiences.
0-39% (0-7)	Generic.

B4: Meridian Wealth (LO1, LO2, LO5)

Most demanding question. Bundles graphical integrity, pre-attentive perception, wrong palette family, ethics, and AI accountability into one integrated critique.

(a) Evaluation of analyst's defence (20 marks)

Band	Descriptor
70%+ (14-20)	All five strands with depth. Data accuracy vs graphical integrity (Tufte; lie factor far above 1.0 for 0.4-pt spread shown as near-vertical staircase). Pre-attentive shape perception (Ware): viewers absorb slope before reading axis labels; "labels are visible" is a fig leaf. Cleveland and McGill: position along common scale should not be distorted. Diverging palette misuse: red-green divides "winners from losers" with no analytical midpoint; sequential or categorical palette required. Ethics: investors are protected stakeholders; misleading visuals carry regulatory exposure (ESMA, MiFID II). AI accountability: tool optimises for output not integrity; "the AI did it" is professional abdication.
60-69% (12-13)	Four of five strands with depth.
50-59% (10-11)	Three strands; "truncated axes mislead" without full perceptual mechanism.
40-49% (8-9)	Two strands at surface level.
0-39% (0-7)	Cannot critique.

(b) QA framework (15 marks)

Band	Descriptor
70%+ (11-15)	Four+ criteria each grounded: (1) Graphical integrity (Tufte; lie factor ~1.0; axis baseline justified or clearly broken); (2) Perceptual channel fit (Cleveland and McGill, position for precise comparison; appropriate palette family); (3) Cognitive clarity (Few; Knaflitz, hierarchy, single dominant message, annotated insight); (4) Ethical and accessibility check (Kirk, attribution, methodology, CVD safe, no selective omission); (5) Audience and context check ("so what" test). Each criterion explained with practical application step.
60-69% (9-10)	Four criteria, most grounded.
50-59% (7-8)	Three criteria, partial grounding.
40-49% (6)	Two criteria, weak grounding.
0-39% (0-5)	Fewer than two; incoherent.

B5: HarbourCity Open Data (LO1, LO2, LO5)

Tests audience-design plus visualisation ethics (data sparsity systematically disadvantages a community) plus the ethical risk of "clean and simple" instructions that suppress legitimate uncertainty.

(a) Three-audience design challenge (15 marks)

Band	Descriptor
70%+ (11-15)	Layered architecture with audience-appropriate entry points. Commuters: fast-loading mobile map, route-level congestion, ETA. Journalists: summary statistics, trend comparisons, exportable data. City planners: historical analysis, intersection-level drill-down, demographic and investment overlays. Shared data model; presentation layer adapts. Shneiderman's mantra; audience-first design (Kirk); progressive disclosure.
60-69% (9-10)	Three audiences differentiated, principles named.
50-59% (7-8)	Three audiences, recommendations overlap.
40-49% (6)	Two audiences or no architecture.
0-39% (0-5)	Single design imposed on all.

(b) Ethics of sparse-sensor underrepresentation (20 marks)

Band	Descriptor
70%+ (14-20)	Frames issue as visualisation ethics, not just data quality. Sensor sparsity is a data limitation; the visualisation claim ("congestion index for HarbourCity") is what overstates what the data can support. Discusses: equity implications of underrepresenting a community; persuasive power of polished public dashboards; "clean and simple" instruction can suppress essential caveats and become an ethical failure. Specific features: confidence indicators or coverage maps shown alongside the index; explicit disclosure of sensor density and known limitations; community input channel; commitment to addressing the gap; accessible methodology page. References to ethical visualisation (Kirk; Knaflitz; D'Ignazio and Klein, <i>Data Feminism</i>) push to top band.
60-69% (12-13)	Engages with both technical and trust dimensions; several specific features.
50-59% (10-11)	Identifies data quality issue but treats primarily as technical fix.
40-49% (8-9)	Surface-level.
0-39% (0-7)	Fails to engage with the ethical dimension.

General Marking Principles

1. **Reward DataVis Challenge engagement.** Specific references (dataset, design choices, audience response, what worked, what would change) are evidence equivalent to theoretical citations.
2. **Named principles valued and expected for top marks.** Applied understanding can reach 60%; 70%+ should demonstrate engagement with named theory, framework, or wider reading.
3. **Reward structure and reasoning** over volume.
4. **Reward honesty and critical reflection.** Acknowledging limitations is Level 9 thinking.
5. **Penalise vagueness, not informality.**
6. **Scenario engagement matters.** Generic answers in B2/B3/B4/B5 capped at 50-59%.
7. **70% threshold:** at least two of (a) accurate use of named theories, (b) connection to applied DataVis Challenge experience, (c) wider reading, (d) genuine critical evaluation.

LO Coverage Matrix

Question	LO1	LO2	LO5
A1	Primary		
A2		Primary	
A3	Primary		Secondary
A4		Primary	Secondary
B1	Primary		Primary
B2		Primary	Primary
B3	Primary	Secondary	Primary
B4	Primary	Primary	Primary
B5	Primary	Primary	Primary

All three examined LOs are assessable through compulsory Section A. Any combination of two Section B answers covers all three LOs multiple times.